

Ventilation & Indoor Air Quality

CO₂ • MVHR • ASHRAE 62.1 • DCV • ACH • IAQ Pollutants

What You Will Learn

- Indoor air pollutants: CO₂, CO, VOCs, PM2.5, humidity, radon — sources and health thresholds
- Three ventilation types: natural, mechanical, and mixed mode — how each works
- ASHRAE 62.1 minimum outdoor air rates and ACH requirements by space type
- MVHR heat recovery ventilation — how it saves 85% of heating energy lost through ventilation

ASHRAE 62.1 — Minimum Outdoor Air (examples)

Office:	8.5 L/s·person + 0.8 L/s·m ²
Classroom:	5 L/s·person + 0.6 L/s·m ²
Hospital ward:	8 L/s·person + 2.5 L/s·m ²

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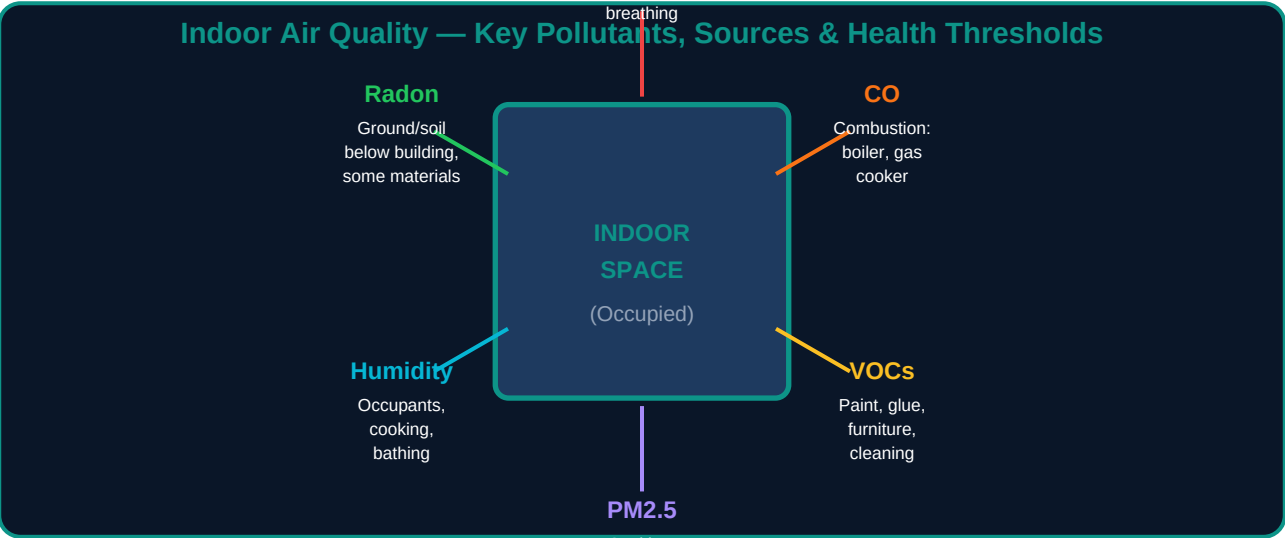
MEP-007: Ventilation & Indoor Air Quality

People spend over 90% of their time indoors — in offices, homes, schools, hospitals, and shopping centres. The quality of the air in those spaces directly affects health, cognitive performance, productivity, and comfort. Studies by Harvard T.H. Chan School of Public Health found that employees in well-ventilated offices scored 101% higher on cognitive function tests than those in conventional offices. Ventilation is not an optional feature of good building design — it is a fundamental health requirement.

Learning Objectives

- Identify the six main indoor air quality pollutants, their sources, and their health thresholds
- Compare natural, mechanical, and mixed-mode ventilation strategies
- Apply ASHRAE 62.1 minimum outdoor air rates to calculate required ventilation for different spaces
- Describe MVHR (heat recovery ventilation) and calculate energy saved versus exhaust-only ventilation
- Design a demand-controlled ventilation (DCV) strategy using CO₂ sensors

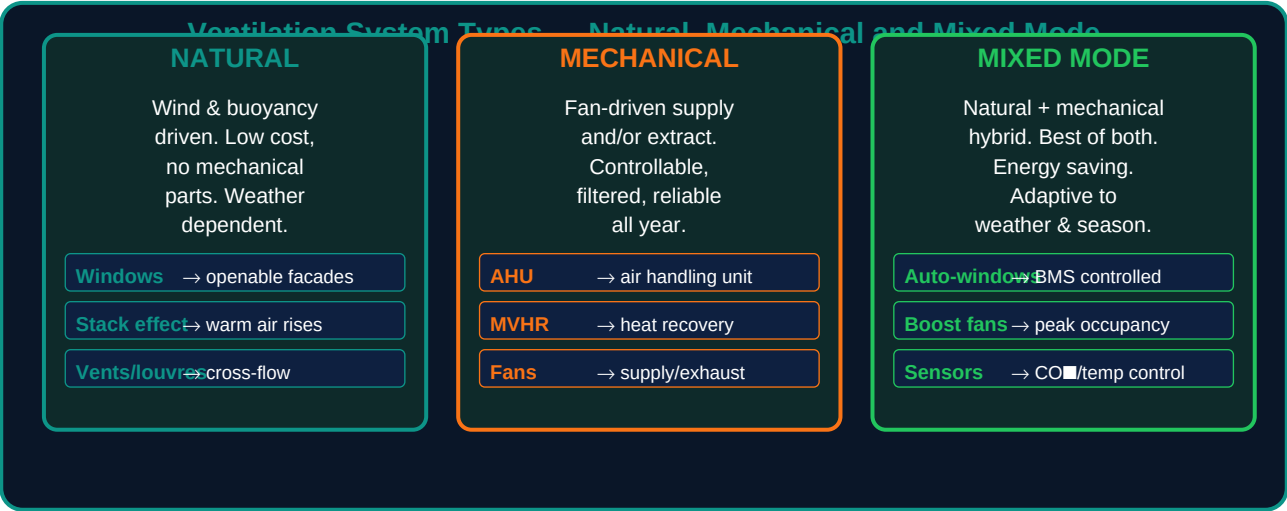
1. Indoor Air Quality — What Pollutes Indoor Air



Pollutant	Primary Sources	WHO / ASHRAE Guideline	Health Effect Above Threshold
CO ₂ (Carbon dioxide)	Occupant breathing: each person produces ~0.2 L/min CO ₂	<1,000 ppm — acceptable. 1,000–2,500 ppm: drowsiness, reduced concentration.	Above 5,000 ppm: headache, nausea. Above 40,000 ppm: life-threatening.

CO (Carbon monoxide)	Incomplete combustion: gas boilers, cookers, paraffin heaters, generators indoors	9 ppm (8-hour TWA). 35 ppm (1-hour max). WHO: 10 mg/m ³ (8h)	Above 70 ppm: headache. Above 150 ppm: unconsciousness. Above 400 ppm: fatal in hours.
VOCs (Volatile Organic Compounds)	Paints, adhesives, new furniture/carpet, cleaning products, air fresheners, printers	WHO: <300 µg/m ³ for total VOCs. Formaldehyde: <0.1 mg/m ³	Short-term: eye/nose irritation, headaches. Long-term: some are carcinogenic (benzene, formaldehyde).
PM2.5 (Fine particulates)	Outdoor air ingress, cooking, candles, smoking, printers, dust resuspension	WHO annual: 5 µg/m ³ . WHO 24h: 15 µg/m ³ . ASHRAE 62.1: 35 µg/m ³	Penetrates deep lung tissue. Linked to cardiovascular disease, lung cancer, premature death.
Humidity (RH)	Occupants, cooking, bathing, wet materials, HVAC system sizing	40–60% RH optimal. Below 30%: skin/mucosa drying. Above 65%: mould growth.	Low RH: increased virus transmission, skin discomfort. High RH: mould, dust mites, structural damage.
Radon (Rn-222)	Natural radioactive gas from granite soils and rocks below buildings	WHO action level: 100 Bq/m ³ . UK action level: 200 Bq/m ³	Long-term exposure: lung cancer risk (second leading cause after smoking in many countries).

2. Ventilation System Types



3. ASHRAE 62.1 — Minimum Ventilation Rates

ASHRAE 62.1 (Ventilation for Acceptable Indoor Air Quality) is the primary international standard for minimum outdoor air ventilation rates in non-residential buildings. It uses the **Ventilation Rate Procedure**: a two-component formula combining a person-dependent component and a floor-area-dependent component. Most building codes worldwide reference ASHRAE 62.1 or an equivalent national standard.

Outdoor Air Rate per Zone = (Rp × Pz) + (Ra × Az)

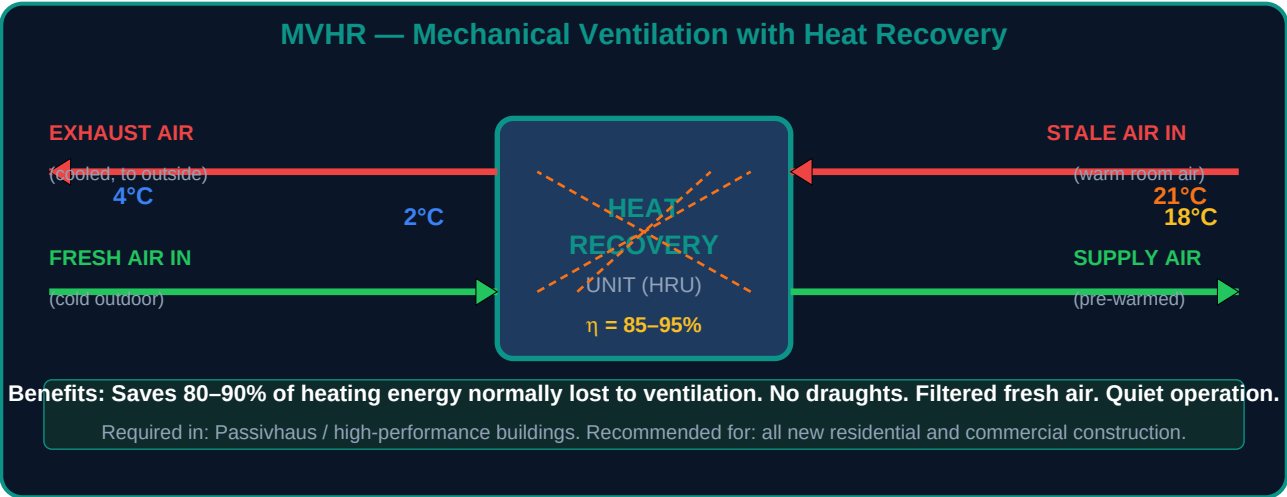
Rp = people outdoor air rate (L/s·person); Pz = number of people in zone;
Ra = area outdoor air rate (L/s·m²); Az = floor area of zone (m²).

Space Type	Rp (L/s·person)	Ra (L/s·m²)	Typical ACH	Notes
Office (general)	8.5	0.8	4–6 ACH	Default occupancy: 5 persons per 100 m². Includes space for equipment heat loads.
Conference room	8.5	0.8	6–10 ACH	Higher occupancy density — calculate actual occupancy, not area-based estimate.
Classroom (school)	5.0	0.6	4–8 ACH	CO ₂ monitoring mandatory in many jurisdictions. High CO ₂ reduces learning performance.
Hospital patient room	8.0	2.5	6 ACH (ANSI/ASHRAE 170)	Minimum 2 ACH outdoor air. Negative pressure for isolation rooms.
Hospital operating room	8.0	2.5	20 ACH minimum	All air must be filtered (HEPA). Positive pressure required. No recirculation.

Retail store	7.5	1.5	4–6 ACH	High occupancy density in sales areas requires careful zoning.
Restaurant dining	15.0	1.5	8–12 ACH	Higher rate for cooking odours and smoke. Separate kitchen exhaust required.
Hotel bedroom	5.0	0.6	2–4 ACH	Minimum rates apply when occupied. Many use fan coil unit + corridor makeup air.

4. MVHR — Mechanical Ventilation with Heat Recovery

MVHR (also called HRV — Heat Recovery Ventilator or ERV — Energy Recovery Ventilator) is the highest-efficiency mechanical ventilation system. It extracts stale air from the building and simultaneously supplies fresh outdoor air, passing both streams through a heat exchanger. Up to 95% of the heat in the exhaust air is transferred to the incoming fresh air before it is exhausted outside — dramatically reducing heating costs compared to any system that simply exhausts warm air to the outdoors.



MVHR vs Alternative	Energy Comparison	When MVHR is Required / Recommended
Extract-only fan (no recovery)	All ventilation heat lost — heating system must replace it all. 100% energy loss on ventilation.	Old buildings, low budget, where simple extract is sufficient code minimum.
Supply + extract without recovery	Both streams run but no energy transfer. Slightly better than extract-only for comfort.	Rarely specified in modern buildings — MVHR is only marginally more expensive.
MVHR (heat recovery only)	Recovers 75–90% of sensible heat. Does not recover latent heat (moisture).	Cold, dry climates. Residential and commercial buildings where winter heating dominates the energy bill.
ERV (energy/enthalpy recovery)	Recovers 70–80% of both sensible (temperature) AND latent (humidity) energy.	Hot/humid climates (tropical, Middle East, East Africa) where dehumidification load is significant.

5. Demand-Controlled Ventilation (DCV)

DCV modulates ventilation rates in real time based on measured occupancy or CO₂ concentration. Instead of ventilating at full design rate whether the space is empty or packed, DCV reduces airflow when the space is unoccupied (saving energy) and increases it when CO₂ rises (maintaining air quality). DCV can save 20–40% of HVAC energy in variable-occupancy spaces such as conference rooms, classrooms, and auditoriums.

DCV Sensor Type	Measures	Control Action	Accuracy / Notes
CO ₂ sensor (NDIR)	Parts per million (ppm) CO ₂ concentration	Increase supply air when CO ₂ > setpoint (typically 1,000 ppm). Decrease when CO ₂ < 800 ppm.	Non-dispersive infrared (NDIR) is the only accurate technology for CO ₂ . Semiconductor sensors are not reliable.
Occupancy sensor (PIR/ultrasonic)	Motion / presence in zone	Reduce to minimum ventilation when no occupancy detected for 10–15 min. Return to design rate immediately on occupancy.	Faster response than CO ₂ . Cannot distinguish 1 vs 50 people — use CO ₂ for occupancy-density control.
VOC sensor	Total VOC concentration (approximation)	Increase ventilation when VOCs detected. Used in kitchens, printing areas, laboratories.	Less accurate than CO ₂ for general occupancy. Useful as a secondary control in areas with chemical sources.

6. Air Changes per Hour (ACH) by Space Type

ACH (Air Changes per Hour) is the number of times the entire volume of air in a space is replaced per hour. It is a simpler way to specify ventilation than the ASHRAE two-component method, widely used for residential and small commercial applications. However, ACH alone does not account for occupancy density — always supplement ACH with a per-person outdoor air check for occupied spaces.

$$Q_{\text{ventilation}} \text{ (m}^3\text{/h)} = \text{ACH} \times \text{Volume (m}^3\text{)}$$

Space / Application	Minimum ACH	Recommended ACH	Source / Notes
Residential bedroom	0.5 ACH	1.0 ACH	UK Building Regs Part F (2021). ASHRAE 62.2 for residential.
Residential kitchen	15 ACH (extract when in use)	30 ACH (extract at cooker hood)	Extract rate, not continuous ventilation. All air must be exhausted externally.
Residential bathroom/WC	6 ACH min	15 ACH (extract)	Extract-only. Minimum run-on after use: 15 minutes.
Office open plan	4 ACH	6–8 ACH	CIBSE Guide A. Supplement with ASHRAE 62.1 person-based check.
Server room / data centre	30–60 ACH	Varies by heat load	ACH depends entirely on heat load, not occupancy. Compute from kW heat rejection.
Hospital isolation (negative pressure)	12 ACH total	12 ACH (6+ ACH outdoor air)	ANSI/ASHRAE/ASHE 170. All air exhausted — no recirculation. 2.5 Pa negative pressure.
Laboratory (fume hoods)	6 ACH min	10–12 ACH	Must be confirmed by fume hood face velocity calculation. ANSI Z9.5.
Car park (basement)	6 ACH	10 ACH	CO monitoring mandatory. Jet fan impulse ventilation or forced extract with CO sensor control.

7. Ventilation System Selection Guide

Application	Recommended System	Key Reason
High-performance residential (Passivhaus)	MVHR (85–95% heat recovery) + airtight envelope	Airtight building requires controlled mechanical ventilation. MVHR recovers heat that natural ventilation would waste.

Standard residential refurbishment	Intermittent extract fans (bathroom/kitchen) + trickle vents in windows	Simple, low cost, code-compliant for most existing residential buildings.
Office (single floor)	Central AHU with supply and extract ductwork + DCV CO ₂ control	Allows filtration, humidity control, and energy savings from DCV in variable-occupancy zones.
Retail/commercial with high ceilings	Displacement ventilation (supply at low level, extract at high level)	Efficient air distribution — air rises naturally as it warms, carrying contaminants to exhaust at ceiling.
Hospital or cleanroom	Dedicated AHU per zone + HEPA filtration + pressure differential control	Infection control requires ultra-clean supply air, strict pressure regimes, and no recirculation.
Naturally ventilated school	Mixed mode: operable windows + CO ₂ -triggered mechanical boost fans	Reduces energy cost versus full mechanical. CO ₂ DCV ensures fresh air when windows are closed.

8. Practice Exercises

#	Exercise	Data / Instructions
1	Outdoor air rate calculation (ASHRAE 62.1)	An open-plan office: floor area 300 m ² , 40 occupants (design maximum). $R_p = 8.5$ L/s·person, $R_a = 0.8$ L/s·m ² . Calculate: (a) person component (L/s), (b) area component (L/s), (c) total minimum outdoor air flow (L/s), (d) convert to m ³ /h.
2	ACH calculation	A server room: 8m × 6m × 3m (height). Heat load = 40 kW. Cooling air supply at 16°C, room setpoint 24°C. Use $Q = P / (\rho \times C_p \times \Delta T)$ where $\rho = 1.2$ kg/m ³ , $C_p = 1,005$ J/kg·K. Calculate: (a) required airflow (m ³ /s), (b) ACH. Is this within the recommended range?
3	MVHR energy saving	A house ventilates at 0.5 ACH. Volume = 250 m ³ . Outdoor temp = 2°C, indoor = 21°C. Without MVHR: heat energy to warm ventilation air (kW) = $\rho \times C_p \times Q_{\text{vent}} \times \Delta T$. With MVHR at 85% efficiency: how much heat is recovered (kW)? Annual saving (assume 5,000 hours at this condition): kWh and cost at £0.25/kWh.
4	DCV setpoint design	A conference room holds 20 people maximum. Outdoor air requirement per person = 8.5 L/s. Design a DCV strategy: (a) what is the maximum ventilation rate (L/s)? (b) At what CO ₂ level should full ventilation activate? (c) What is the minimum ventilation rate for an empty room (use R_a only, area = 40 m ²)?
5	IAQ assessment	A newly renovated open-plan office has occupant complaints: headaches, drowsiness, eye irritation. CO ₂ measured at 1,800 ppm. VOC sensor reading elevated. Fresh air grilles are partially blocked. Identify: (a) probable cause of each symptom, (b) immediate remedial actions, (c) long-term design changes to prevent recurrence.

Lesson Summary

Key Concept	Summary Rule
CO ₂ as proxy for occupancy	1,000 ppm = acceptable limit. Above 1,200 ppm = inadequate ventilation. Use CO ₂ sensors for DCV control.
ASHRAE 62.1 formula	$Q_{\text{OA}} = (R_p \times P_z) + (R_a \times A_z)$. Always apply both components — neither person nor area rate alone is sufficient.
ACH calculation	$Q_{\text{vent}} \text{ (m}^3\text{/h)} = \text{ACH} \times \text{Room Volume (m}^3\text{)}$. ACH alone is not sufficient for occupied spaces — check per-person rate too.
MVHR efficiency	Recovers 75–95% of sensible heat. Required in airtight buildings. ERV also recovers latent heat (humidity) — use in humid climates.
DCV principle	Modulate ventilation based on CO ₂ or occupancy sensor. Saves 20–40% HVAC energy in variable-occupancy spaces.
Natural vs mechanical	Natural: low cost, weather-dependent, no filtration. Mechanical: controllable, filtered, energy-intensive. Mixed mode: best balance.

Negative pressure rooms	Hospital isolation rooms: minimum 12 ACH total, 6+ ACH outdoor, 2.5 Pa negative. Prevents airborne infection spread.
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What Is Next — MEP-008

MEP-008: Ductwork Design and Sizing — Duct shapes (rectangular, circular, oval), duct materials (galvanised steel, flexible, fabric), friction loss method, equal friction method, velocity reduction method. Calculating duct sizes from airflow requirements. Static pressure and fan selection. Duct insulation and acoustic performance.

Resource	Details
MEP PDF Series	MEP-001 through MEP-030 — ayetechub.com/pdfs.html
HVAC Calculator	Free load calculation tools — ayetechub.com/downloads
Community	Telegram: t.me/ayetechub